

Scott J. Gilbert

Schaumburg, IL | hello@scottgilbert.dev | <https://linkedin.com/in/scott-j-gilbert> | <https://scottgilbert.dev> | <https://github.com/ScottJGilbert>

Aspiring computer engineer with a strong foundation in hardware design, embedded systems, and software development. Skilled in C, C++, Python, embedded systems, and data processing, with hands-on experience in real-time systems, machine learning, CI/CD, and web development.

Education

University of Illinois Urbana-Champaign

GPA: 4.00/4.00

Activities and Societies: Campus Honors Program, Illini Solar Car, Project: Code, Illini Redstone Computing, Engineers without Borders, Marching Illini
Relevant Coursework: Calculus III (MATH 241), Intro Differential Equations (MATH 285), Intro to Computing (ECE 120), Computational Linear Algebra (MATH 257), Intro to Electronics (ECE 110), Computers & Programming (ECE 220), Quantum Physics (PHYS 214), Thermodynamics (PHYS 213)

B.S., Computer Engineering, May 2029

Grainger College of Engineering

James B. Conant High School

High School Diploma, May 2025

Weighted GPA: 4.92/4.00 | **Unweighted GPA:** 4.00/4.00

Activities: Congressional Debate, Marching Band, Jazz Band, Honor Band/Orchestra, Cross Country, Track and Field, National Honor Society

Relevant Coursework: AP Calculus BC, AP Computer Science Principles, AP Computer Science A, AP Physics C

Professional Experience

Computer Administrator

October 2025 - Present

University of Illinois Campus Honors Program (CHP)

Urbana, Illinois

- Maintain CHP's computing hardware infrastructure, integrations with external storage and web services.
- Collaborate with university admissions, IT services, and human resources to process thousands of data points across pipelines in a timely manner.

Full-Stack/AI Intern

June 2025 - August 2025

Team2Go Inc.

Yongin-si, Gyeonggi, South Korea (Remote)

- Engineered Dockerized systems for internal and external-facing tools, enhancing integration between backend applications and user interfaces.
- Integrated custom OpenAI GPT models with streaming output, document-based vector embedding, and rigid boundary/instruction declaration.
- Collaborated with executives through regular progress updates and delivered projects ahead of schedule while exploring LLM design/psychology.

Leadership/Involvement

Electrical Team/Array Team Member

Illini Solar Car | Oct 2025 - Present

- Develop maximum power point tracking firmware and solar array layout specifications, contributing to overall electrical and vehicle optimization.
- Participate in battery characterization, public showcases/vehicle demonstrations, composites fabrication, and other electrical/general car work.
- Hold core leadership positions as Webmaster, Array/MPPT lead, and lead radio technician for 2026-2027.

President & Systems Administrator

Illini Redstone Computing | Sep 2025 - Present

- Manage day-to-day financial, administrative, and technical decision-making while acting as primary liaison to the university/other organizations.
- Develop and maintain a network of containerized applications across gaming, authentication, user management, and data pipelining services.

Backend Team Lead

Project: Code UIUC | Sep 2025 - Dec 2025

- Managed team focused on deploying data handling, APIs, and core logic for an data-driven, all-in-one crop monitoring and plant health solution.
- Held weekly meetings, maintained production software, specified versioning and CI systems, track progress, and reported progress to project lead.

Eagle Scout

Scouting America | Mar 2018 - Dec 2024

- Planned and organized a coordinated trail restoration effort where volunteers contributed over 100 service hours.
- Collaborated with sponsors to secure necessary resources/tooling to finish the project a week ahead of schedule with zero workplace incidents.

Projects

Personal Content System

May 2025 - Present

- Developing flexible microservices-based architectures integrating a portfolio, blog, authentication, account management, AI-driven content development, workflow automation & CI/CD tooling and service integration, and a rich, data-driven backend through container orchestration.
- Modernizing backend with scalable technologies like GraphQL for improving performance and expanding application and cross-platform use cases.

Illinois Congressional Debate Association Website

May 2024 - August 2025

- Led development of a full-stack application, improving front-end utility with React and performance/scalability with a backend service and CMS.
- Programmed, tested, and deployed features improving mobile accessibility, information flow, site security, and page/component load times.
- Achieved strong search visibility, reaching ~20K impressions/6% CTR over four ranking among Google's top ten student congress sites.

Solar Energy Educational Demonstration

October 2024 - April 2025

- Developed Arduino-based systems for temperature data acquisition, with custom firmware to interface with real-time visualization tools.
- Created interactive demonstrations, integrated computing elements, and technical infographics to teach solar concepts and energy efficiency.
- Collaborated with event organizers to deploy demonstrations for children and families at a community Earth Day event.

Skills

- **Programming:** C, C++, Python, SQL, Java, Assembly, TypeScript, PHP, Wolfram
- **Databases:** MySQL, SQLite, PostgreSQL, pgvector, S3, Uploadthing, Backblaze
- **Numerical Analysis:** Jupyter Notebook, NumPy, SciPy, Matplotlib, Mathematica, Excel, Google Sheets
- **Embedded Systems/Microcontrollers:** Arduino, ELEGOO, KiCad, NXP
- **Industry Tools:** Linux, Docker, Docker Compose, Xilinx Vivado, MCUxpresso, LTSpice, Git, GitHub, Google Workspace, Microsoft Office, Slack
- **Workflow Automation & CI/CD:** Vercel, Claude Code, GitHub Copilot, GitHub Actions, n8n, Make
- **Technical Expertise:** Firmware Design, Backend Development, PCB Design, AI/ML Application, Virtualization, Object-Oriented Programming

Awards: Dean's List, National Merit Scholarship, AP Scholar with Distinction, Illinois State Seal of Biliteracy, Gold President's Volunteer Service Award